

Introduction to Computer Organisation and Architecture UCCD1133



Chapter 3 Basic Concept of Logic

Disclaimer



- This slide may contain copyrighted material of which has not been specifically authorized by the copyright owner. The use of copyrighted materials are solely for educational purpose. If you wish to use this copyrighted material for other purposes, you must first obtain permission from the original copyright owner.

Outline

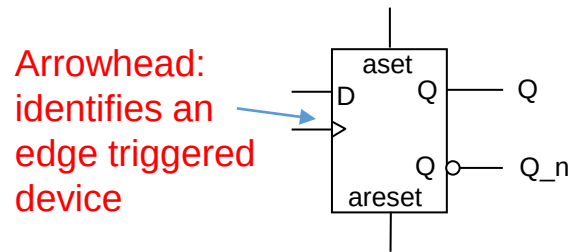
- Memory elements
 - Flip flop and latch
 - register

Memory elements

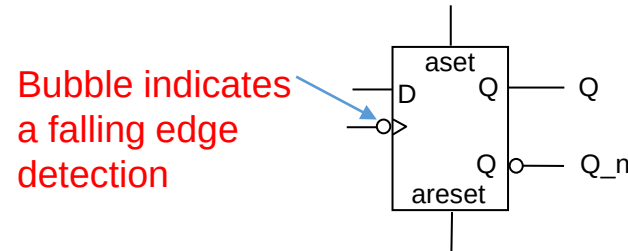
- Memory elements are the most primitive sequential circuits for storing a bit value.
- 2 types of memory elements
 - i. Latch (clockless)
 - ii. Flip-flop (clock based)
- Common types of flip-flop:
 - i. Set-Reset flip-flop (SRFF).
 - ii. JK flip-flop (JKFF).
 - iii. Delay flip-flop (DFF)
 - iv. Toggle flip-flop (TFF).

Delay flip-flop (D-FF)

- A flip-flop has edge sensitive behavior.
 - The output responds to the input data only during the active edge of the clock.
- Logic symbols of DFF:

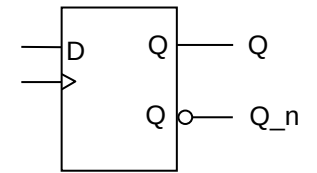


(a) Positive-edge triggered DFF



(b) Negative-edge triggered DFF

- The arrowhead pin: usually apply periodic signal – e.g. **clock signal**
- One FF -> one clock pin

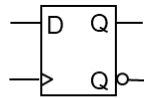


(c) No aset and areset input

- Aset: output become 1; areset: output become 0. (asynchronous inputs – regardless of clock).
- When not used, aset and areset inputs no need to show. Its only logic symbol.
- But in actual hardware implementation, need to connect them to values that de-activate them.

Delay flip-flop (D-FF)

- A DFF copies input value at D (from one cycle) to output Q (in the next cycle) when active clock edge occurs.
- At other times, not copied.



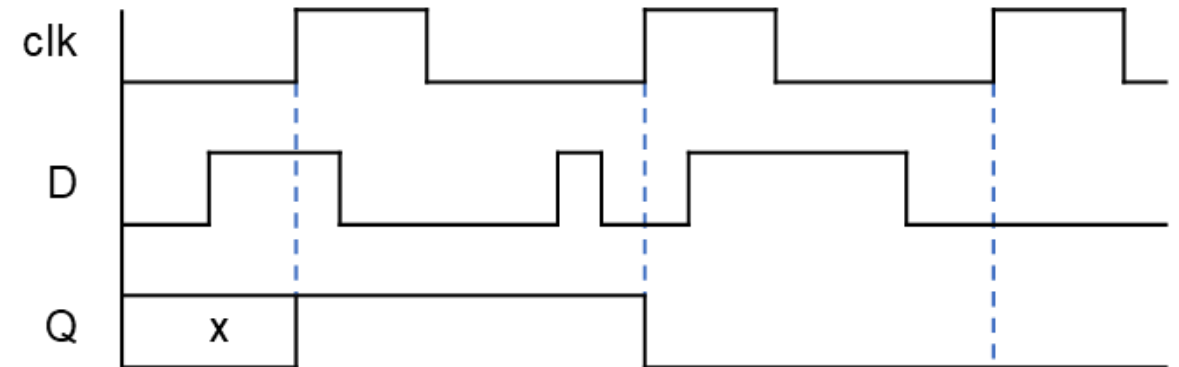
(a) DFF logic symbol

Inputs	Output	Function
D	Q*	
0	0	Copy
1	1	Copy

(b) Function table

- 3 basic functionalities
 - To synchronise a signal to clock
How? Copy value at D to Q only during the active edge of the clock.
 - To delay a signal at D by one cycle.
The value should appear at Q in the next cycle (relative to the time at D).
 - To filter glitches that appears in a signal

D flip-flop copies D to Q on the rising edge of the clock, and remembers its state at all other times



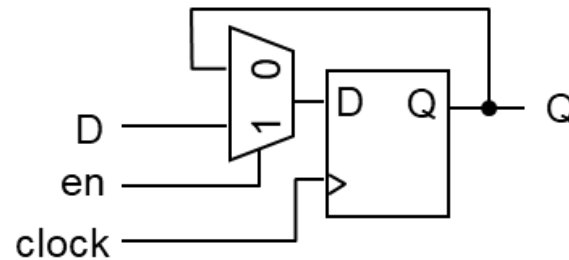
(c) Timing diagram (rising/positive edge triggered)

DFF with enable

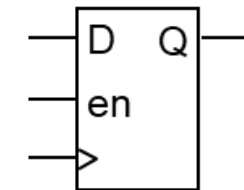
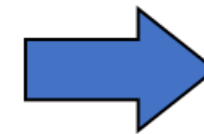
- However, the previous DFF cannot store or hold a signal value.
- To hold a value, need to add a hold function into DFF
 - ✓ A DFF with enable

Inputs		Outputs	Function
Enable input E	D	Q*	
0	d	Q	Hold
1	0	0	Copy
1	1	1	Copy

Function table for DFF with enable



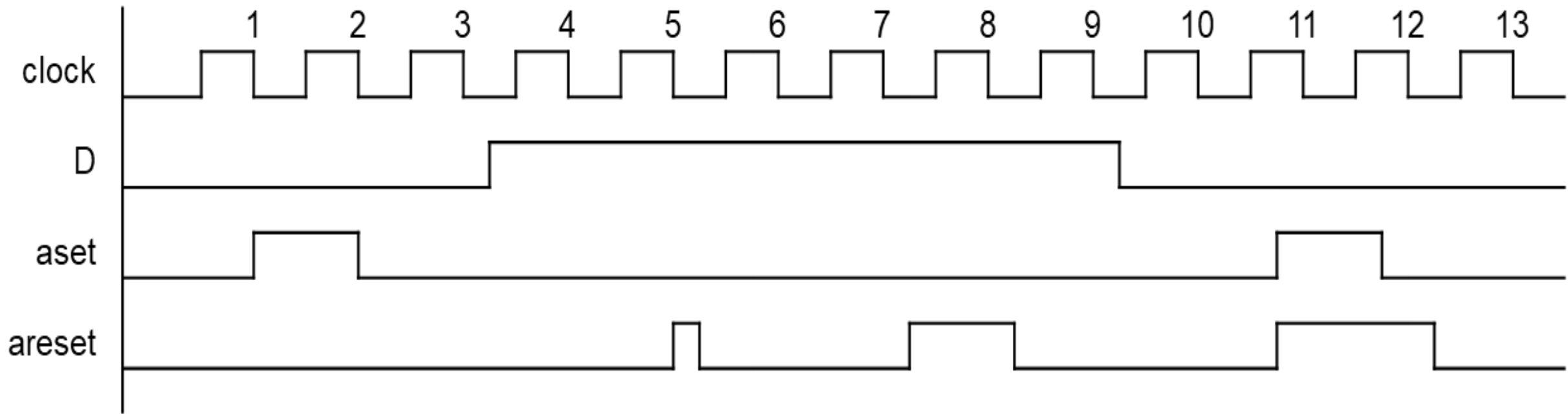
(a) Creating a storage effect by adding a Hold function into a DFF



(b) DFF with enable

Delay flip-flop (D-FF)

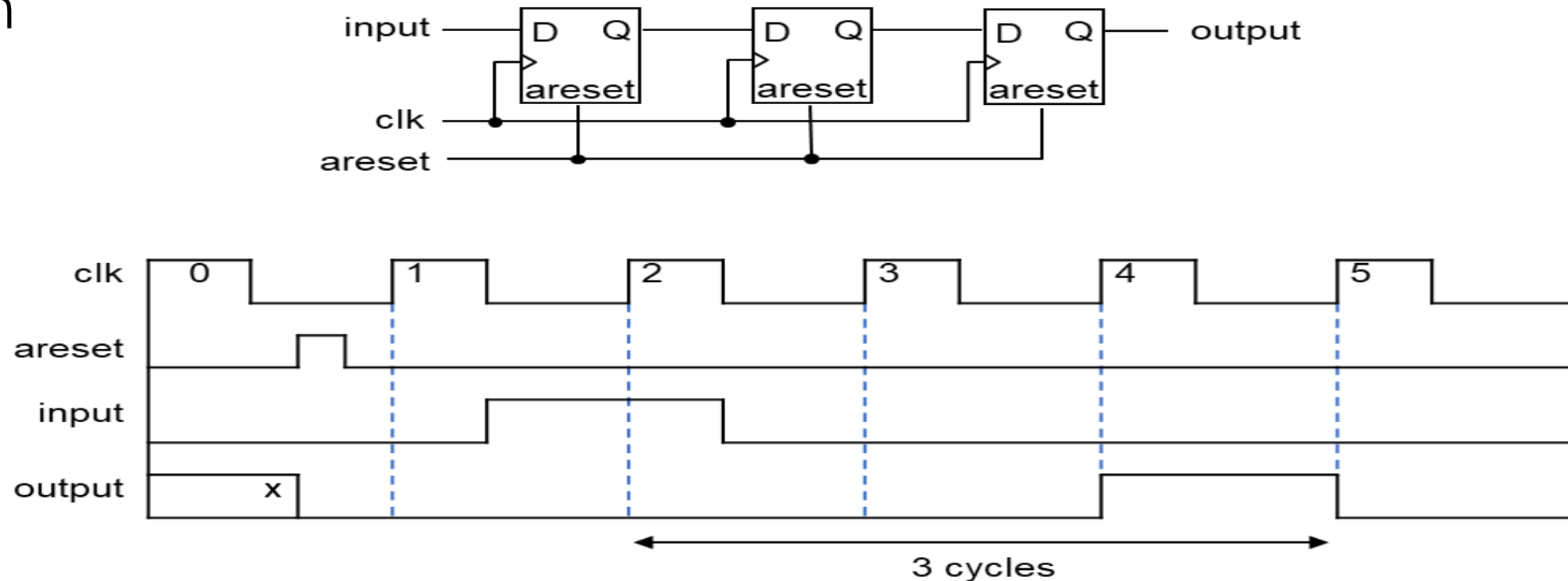
Example 15: Draw the output waveform of the positive-edge DFF given the input signals as below



Delay flip-flop (D-FF)

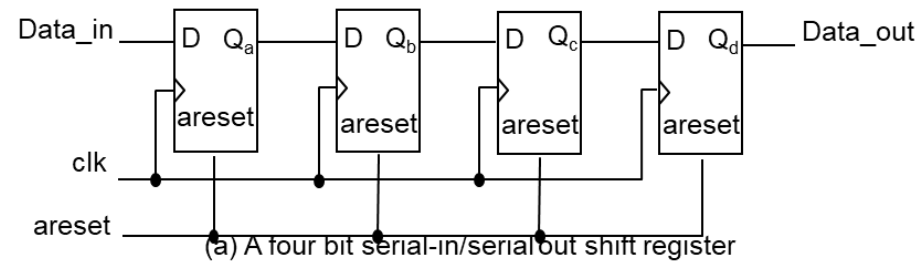
Example 16: Show how DFFs can be used to delay a signal by 3 clock cycles. Draw a timing diagram to illustrate the delay.

Solution



Registers

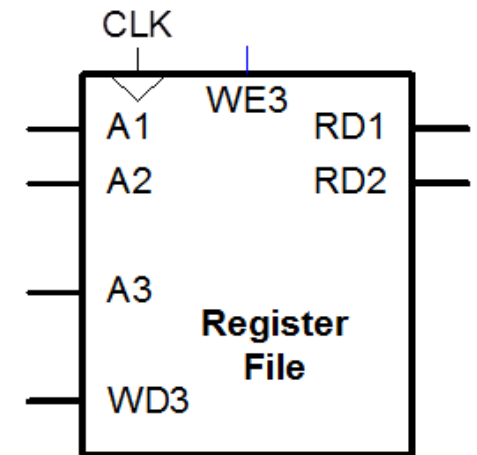
- Register consists of a set of N flip-flops with a common clock input.
- N is typically an integral power of 2
- It is mainly build from D-type flip-flops

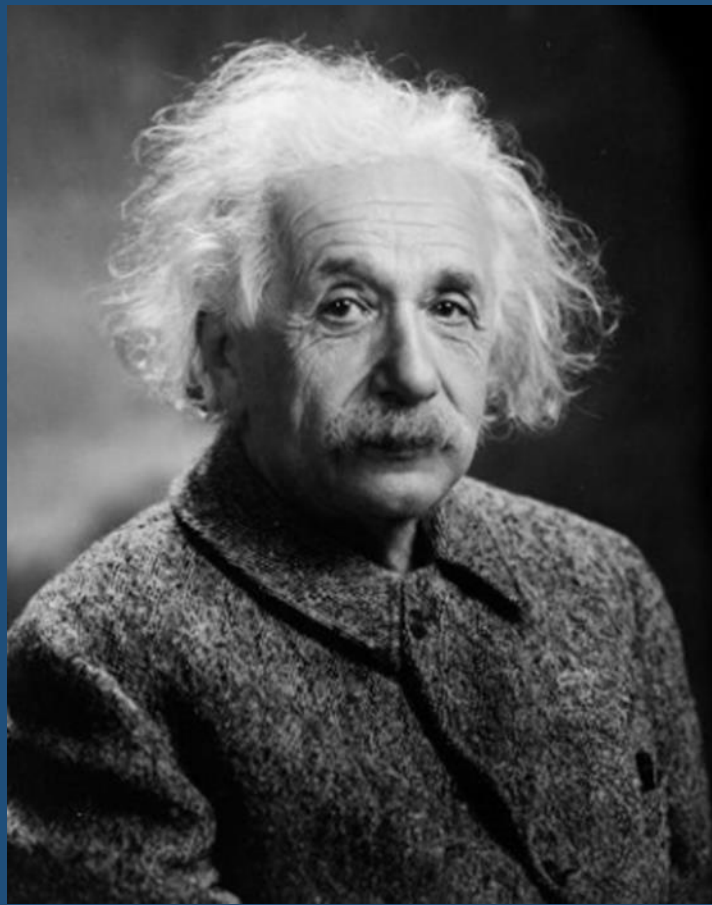


- Registers can be replicated to build larger memory units. It is typically used as a fast temporary storage in a central processing unit (CPU).
- Each flip-flop can store one bit of information; therefore, the number of stages (flip-flops) in a register determines its storage capacity.
- A typical CPUs either store 32 or 64 bits in registers.

Registers File

- A register file is a collection of registers.
 - ✓ Each of it is individually addressed.
 - ✓ By specifying the register number to be accessed, the register file can be read and written.
- Registers are grouped in array to form $M \times N$ register file
 - ✓ M registers, each N -bit wide
 - ✓ E.g. 4×32 dual-port
 - ✓ Four 32-bit registers
- The interface of a register file should at least has:
 - ✓ A clock signal (CLK)
 - ✓ An input to specify the register number (address) (A)
 - ✓ An output read port (RD)
 - ✓ An input write port (WD)
 - ✓ A control bit(s) to enable/disable write operation (WE)





“Creativity is
intelligence
having fun”

– Albert Einstein



Q & A

Thank you